The Solar System

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Written by: Ashika
Reviewed by: Caroline Carroll
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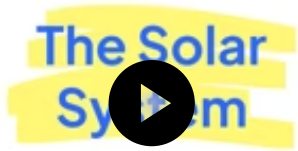

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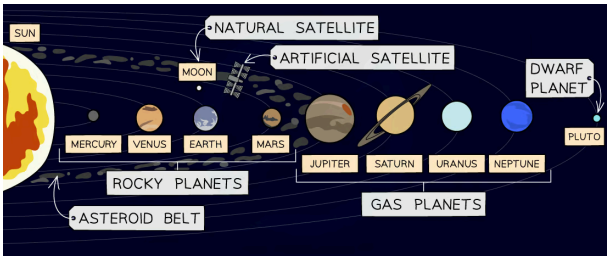
The Solar System

- The Solar System consists of:
  - the Sun
  - eight planets
  - natural and artificial satellites
  - dwarf planets
  - asteroids and comets

The Solar System



The Solar System consists of one star (the Sun) and the objects that orbit it, including the planets, moons, dwarf planets, asteroids and comets

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

The Planets

- There are **eight** planets which **orbit** the Sun

Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune

- The planets can be divided into two groups
  - the **inner** rocky planets
  - the **outer** gas giants

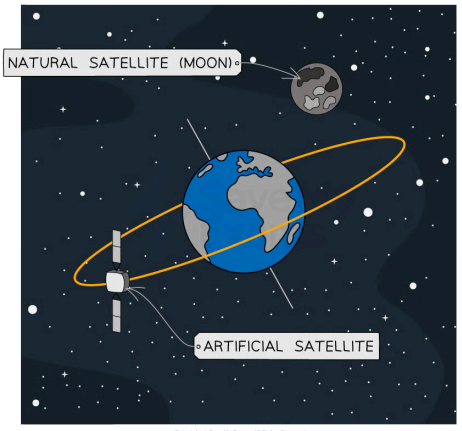
Dwarf Planets

- A dwarf planet is an object similar to a planet, but much smaller
- The **gravitational field** around a planet is strong enough to pull in nearby objects (with the exception of natural satellites)
- Whereas, the gravitational field around a dwarf planet is **not strong enough** to pull in nearby objects

Satellites

- There are two types of satellites: **natural** and **artificial**

Natural and artificial satellites of Earth



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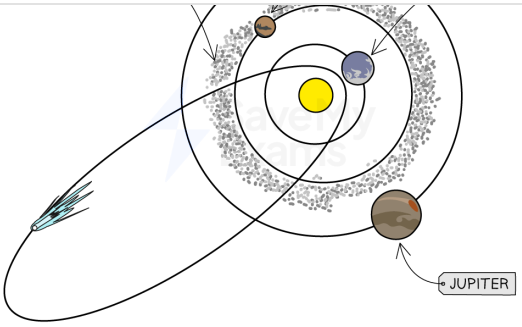
The Moon is a natural satellite of the Earth. Many artificial satellites orbit around the Earth.

- Natural satellites are objects that orbit planets
- A **moon** is a type of natural satellite
- **Artificial satellites** are manmade objects that orbit another object in space
- The International Space Station (ISS) is an example of an artificial satellite that orbits the Earth

Asteroids and comets

- Asteroids and comets also orbit the Sun

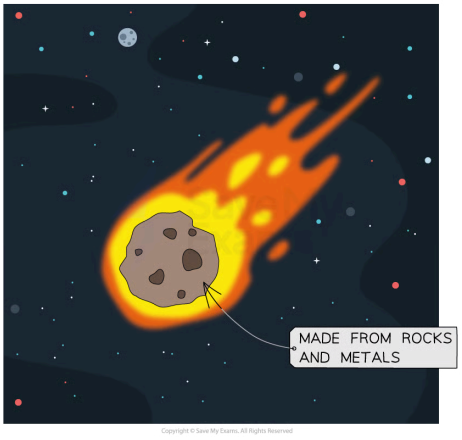
Locations of asteroids and comets



Asteroids are found in the Asteroid Belt between Mars and Jupiter, whereas comets are usually found in the outer reaches of the Solar System due to their highly elliptical orbits

- An **asteroid** is a small **rocky object** which orbits the Sun
- The **asteroid belt** lies between Mars and Jupiter

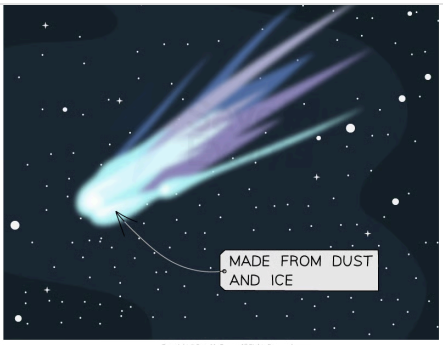
An asteroid



Asteroids are small, rocky objects which occupy the inner Solar System

- A **comet** is an object made of **dust and ice** which goes around the Sun in a highly elliptical (not a circular path with the Sun at the centre) path
- The ice melts when the comet approaches the Sun and forms the comet's **tail**

A comet



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Comets are small, icy objects which occupy the outer Solar System

Orbiting bodies

- The Solar System is made up of many bodies which orbit around other bodies
- **Smaller** bodies **orbit** around **larger** bodies
  - For example, planets orbit the Sun and moons orbit planets
- The orbiting bodies in the Solar System are shown in the table below:

Table of orbiting bodies in the Solar System

orbiting body	body it orbits
planet	the Sun
moon	planet
comet	can pass around the Sun
asteroid	the Sun
artificial satellites	the Earth

Examiner Tips and Tricks

You need to know the order of the 8 planets in the Solar System. The following mnemonic gives the first letter of each of the planets to help you recall them:

My **V**ery **E**xcellent **M**other **J**ust **S**erved **U**s **N**oodles

Mercury, **V**enus, **E**arth, **M**ars, **J**upiter, **S**aturn, **U**ranus, **N**eptune

Extended tier only

- Over many years, data for the planets, moons and the Sun have been collected
- Some of this data is shown in the table below

Data for the planets in the Solar System

Planet	Mean distance from Sun (relative to Earth)	Orbital period (Earth years)	Mean surface temperature (°C)	Density (kg/m^3)	Dia (1
Mercury	0.39	0.24	350	5429	
Venus	0.72	0.60	460	5243	
Earth	1	1	20	5514	
Mars	1.5	2	−23	3934	
Jupiter	5.2	12	−120	1326	
Saturn	9.6	30	−180	687	
Uranus	19	84	−210	1270	
Neptune	30	160	−220	1638	

- The data allows us to
  - make comparisons
  - identify trends and anomalies
  - make predictions
- Some examples of **comparisons** are:
  - Neptune is 30 times **further away** from the Sun than the Earth
  - Jupiter contains the **same** mass as 320 Earths
- An example of a **trend** is:
  - As the distance from the Sun **increases**, the time it takes to complete one orbit (orbital period) also **increases**
- An example of an **anomaly** is:
  - As the distance from the Sun increases, the temperature decreases, **except** for Venus which has a higher temperature than Mercury
- An example of a **prediction** is:
  - The temperature of a dwarf planet in the asteroid belt is **likely** to be around −100°C, but it could be

Examiner Tips and Tricks

Don't panic when you see the large table of numbers - you don't need to memorise any of it, but you need to be able to analyse and interpret it confidently. Look for trends such as one variable increasing whilst the other decreases (or increases). Think carefully about why that may be with what you have already learnt about the planets from this topic. For example, what is the planet made of? What is its distance from the Sun and how does this affect it?

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